

SOC LOG ANALYSIS PROJECT

Security Operations Center - Project

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Threat Detection & Log Analysis Using Splunk SIEM

Field	Detail
Tool	Splunk Enterprise 10.2.0 (Free Tier)
Log Source	Apache Web Server - access_combined format
Index	soc_log
Host	DESKTOP-GT1182G
Total Events	112 log entries
Queries Executed	9 SPL queries
Date	19 February 2026
Difficulty	Beginner / Entry-Level SOC

1. Project Introduction & Motivation

1.1 What is SOC?

A Security Operations Center (SOC) is a centralized function where cybersecurity analysts monitor, detect, analyse and respond to cybersecurity incidents around the clock. They make use of SIEM (Security Information and Event Management) platform which is used to ingest logs from servers, firewalls and endpoints then query them to find patterns that indicate attacks.

This project simulates the core daily workflow of a Tier 1 SOC analyst; ingesting web server logs, writing SPL (Search Processing Language) queries, identifying suspicious behaviour and producing a formal incident report. Every skill practiced here maps directly to real SOC job requirements.

1.2 Why Splunk?

Splunk is the industry-standard SIEM used by enterprises, government agencies and MSSPs worldwide. It was chosen for this project because:

- Free tier available - 500MB/day ingestion at no cost, more than enough for learning.
- Runs natively on Windows - no virtual machine or Linux required.
- SPL is beginner-friendly - similar to SQL with powerful security use cases.
- Industry-recognised tool.
- Used in real SOC environments.

1.3 Project Objectives

- Install and configure Splunk Enterprise on a Windows host machine.
- Ingest Apache web server access logs using a fake log generator.
- Write 9 SPL queries ranging from basic filtering to advanced statistical analysis.
- Identify attack patterns: API targeting, DELETE method abuse, distributed IP scanning.
- Produce a formal incident report with risk ratings and recommendations.
- Document dashboard build steps for ongoing monitoring.
- Publish to GitHub as a complete beginner portfolio project.

1.4 Skills Demonstrated

Skill	Tool/Technique	Job Relevance
Log ingestion & indexing	Splunk Add Data wizard	SOC Analyst Tier 1
SPL query writing	9 search queries	Blue Team Analyst
Threat pattern detection	Status codes, HTTP methods	Incident Responder
Statistical analysis	stats, top, table commands	Security Engineer
Incident documentation	Formal report	Any security role
Dashboard creation	Splunk Dashboard Studio	SOC Analyst

2. Environment Setup

2.1 Hardware

Component	Specification
Device	DESKTOP-GT1182G
OS	Windows 10/11 (64-bit)
Processor	Intel Pentium P6200 @ 2.13 GHz
RAM	4.00 GB (3.68 GB usable)
Storage	256 GB SSD

2.2 Tools Used

Tool	Version	Cost	Purpose
Splunk Enterprise	10.2.0	Free (500MB/day)	SIEM - log ingestion & analysis
Fake Log Generator	Online tool	Free	Generate realistic Apache access logs
Web Browser	Any modern	Free	Access Splunk Web at localhost:8000

2.3 Splunk Installation Steps

1. Downloaded Splunk Enterprise.msi installer from splunk.com (free account required).
2. Ran the .msi installer on Windows; accepted license, created admin credentials.
3. Installed to C:\Program Files\Splunk.
4. Splunk Web launched automatically at <http://localhost:8000>.
5. Logged in with admin credentials created during setup.

2.4 Log Ingestion — Step by Step

6. Navigated to: <http://localhost:8000/en-US/app/search/adddata>
7. Selected Upload, uploaded file directly from local machine
8. Selected file: generated_log_2026-02-18T11_50_10.343Z.txt
9. Set Source Type to: access_combined (Apache log format — auto-extracts all fields)
10. Set Index to: soc_log (custom index created for this project)
11. Clicked Review then Submit - received 'File uploaded successfully' confirmation
12. Verified ingestion: ran `index=soc_log` in search bar - 112 events confirmed

2.5 Log Data Overview

The fake log generator produced realistic Apache web server access_combined format logs. This is the standard Apache format used on real production web servers, making it ideal for SOC training. Each event contains the following auto-extracted fields:

Field	Example Value	What It Tells a SOC Analyst
clientip	219.54.21.207	Source IP of the request - used to identify attackers
method	DELETE	HTTP method - DELETE/PUT are red flags
uri/ uri_path	/api/users	What resource was targeted
status	500	Server response - 400s/500s indicate errors or attacks
bytes	9245	Response size - anomalies can indicate data exfiltration
referer	https://www.example.com	Where the request originated from
useragent	Mozilla/5.0 ...	Browser or tool - automated tools look different

Total events ingested: 112 entries. Time range: 18 - 25 February 2026. Source: generated_log_2026-02-18T11_50_10.343Z.txt

3. SPL Queries & Findings

All queries were run with Time Range set to All Time. The index used is soc_log throughout. Queries are presented in the exact order executed, from basic filters to advanced statistical analysis.

Query 1 - Server Error Detection (Status 500)

Purpose

Identify all HTTP 500 server errors. These are the highest priority alert - they mean the server crashed or failed while processing a request, which can indicate successful injection attacks, resource exhaustion, or API disruption.

SPL Command

```
index=soc_log status=500
```

Results: 14 events

Key events found in the results:

Client IP	Method	URI	Status	Significance
219.54.21.207	DELETE	/api/users	500	CRITICAL — attacker hit the user data API
192.168.227.151	POST	/home	500	Suspicious POST causing server error
10.105.124.73	GET	/products/category/electronics	500	Possible parameter injection
178.131.218.134	GET	/cart	500	Cart endpoint crashing — possible exploit
172.17.241.72	DELETE	/blog	500	DELETE on blog endpoint — destructive intent
132.106.20.21	PUT	/cart	500	PUT method causing crash
10.74.75.54	GET	/checkout	500	Checkout endpoint error — business impact
134.147.251.66	GET	/login	500	Login endpoint error — auth disruption

Search | Splunk 10.2.0

localhost:8000/en-US/app/search/search?q=search%20index%3Dsoc_log%20status%3D500&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ra...

New Search

index=soc_log status=500

Time range: All time

14 events (before 2/19/26 10:27:58.000 AM) No Event Sampling

Job

Events (14) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 millisecond per column

Format Show: 20 Per Page View: List

	i	Time	Event
>	2/19/26 10:24:49.000 AM	192.168.100.123 - - [Feb 21 2026, 04:08:18] "POST /checkout?sort-price_desc HTTP/1.1" 500 8455 "https://www.twitter.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	31.38.181.47 - - [Feb 19 2026, 04:47:01] "GET /register HTTP/1.1" 500 3806 "https://www.facebook.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.41.14.57 - - [Feb 22 2026, 02:44:13] "PUT /login HTTP/1.1" 500 5079 "https://www.bing.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.18.155.1 - - [Feb 24 2026, 14:41:58] "POST / HTTP/1.1" 500 3942 "https://www.facebook.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
bytes 14
a clientip 14
a file 9
a ident 1
a index 1
linecount 1
a method 4
page 1

Search | Splunk 10.2.0

localhost:8000/en-US/app/search/search?q=search%20index%3Dsoc_log%20status%3D500&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ra...

Hide Fields All Fields Format Show: 20 Per Page View: List

	i	Time	Event
>	2/19/26 10:24:49.000 AM	132.106.20.21 - - [Feb 25 2026, 09:22:30] "PUT /cart HTTP/1.1" 500 6812 "https://www.bing.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.74.75.54 - - [Feb 25 2026, 04:45:09] "GET /checkout HTTP/1.1" 500 4438 "https://www.example.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	134.147.251.66 - - [Feb 21 2026, 16:24:05] "GET /login?category=electronics&q=search+term HTTP/1.1" 500 1645 "https://www.example.com" "Mozilla/5.0 (iPhone; CPU iPhone OS 14_6 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Mobile/15E148 Safari/604.1"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	131.94.141.46 - - [Feb 20 2026, 03:07:12] "GET /cart HTTP/1.1" 500 10779 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	219.54.21.207 - - [Feb 21 2026, 08:34:43] "DELETE /api/users HTTP/1.1" 500 9245 "https://www.example.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	192.168.227.151 - - [Feb 22 2026, 05:16:59] "POST /home?page=2&sort-price_desc HTTP/1.1" 500 5259 "https://www.google.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.105.124.73 - - [Feb 22 2026, 22:12:43] "GET /products/category/electronics?sort-price_desc&page=2&q=search+term HTTP/1.1" 500 5652 "https://www.bing.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0"	
			host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined

a method 4
page 1
a punct 12
a q 1
a referer 6
a referer_domain 5
a req_time 14
a sort 1
a splunk_server 1
status 1
a timestamp 1
a url 12
a url_path 10
a url_query 5
a user 1
a useragent 5
a version 1

2 more fields
+ Extract New Fields

i	Time	Event
>	2/19/26 10:24:49.000 AM	131.94.141.46 - - [Feb 20 2026, 03:07:12] "GET /cart HTTP/1.1" 500 10779 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	219.54.21.207 - - [Feb 21 2026, 08:34:43] "DELETE /api/users HTTP/1.1" 500 9245 "https://www.example.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	192.168.227.151 - - [Feb 22 2026, 05:16:59] "POST /home?page=2&sort=price_desc HTTP/1.1" 500 5259 "https://www.google.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.105.124.73 - - [Feb 22 2026, 22:12:43] "GET /products/category/electronics?sort=price_desc&page=2&q=search+term HTTP/1.1" 500 5652 "https://www.bing.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.21.214.195 - - [Feb 18 2026, 15:52:43] "GET /contact?page=2&q=search+term HTTP/1.1" 500 8523 "https://www.google.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	178.131.218.134 - - [Feb 20 2026, 09:40:17] "GET /cart HTTP/1.1" 500 5892 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 14_6 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Mobile/15E148 Safari/604.1" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.17.241.72 - - [Feb 19 2026, 04:47:21] "DELETE /blog HTTP/1.1" 500 3335 "https://www.example.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined

SOC Analyst Interpretation

14 out of 112 total events (12.5%) are server errors, this is extremely high for a healthy web server where 500 errors should be less than 1% of traffic. The most critical finding is IP 219.54.21.207 sending a DELETE /api/users request that returned 500, meaning the attacker may have partially disrupted the user data API. Multiple different IPs generating 500s across different endpoints indicates a coordinated, multi-vector attack rather than a single misconfigured client.

Risk Level: **CRITICAL**

Query 2 - Forbidden Access Detection (Status 403)

Purpose

Find all requests blocked by the server with a 403 Forbidden response. These are blocked attack attempts - the server's access controls worked but the IPs should be identified and monitored for escalation.

SPL Command

```
index=soc_log status=403
```

Splunk 10.2.0 interface showing search results for the query `index=soc_log status=403`. The search results are displayed in a table format, showing 7 events. The interface includes a search bar, filters, and a table of results.

Search Query: `index=soc_log status=403`

Time range: All time

Events (7):

Time	Event
2/19/26 10:24:49.000 AM	187.82.139.112 - - [Feb 25 2026, 10:48:11] "GET /services HTTP/1.1" 403 5544 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	35.171.162.120 - - [Feb 25 2026, 03:42:56] "POST /products/category/electronics HTTP/1.1" 403 3793 "https://www.google.com" Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	215.0.26.108 - - [Feb 22 2026, 02:29:44] "GET /api/users HTTP/1.1" 403 1854 "https://www.facebook.com" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	192.168.188.157 - - [Feb 21 2026, 12:52:33] "POST /products/category/electronics HTTP/1.1" 403 4385 "https://www.bing.com" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	172.24.71.97 - - [Feb 25 2026, 01:23:17] "GET /contact HTTP/1.1" 403 8196 "https://www.bing.com" Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	10.222.8.143 - - [Feb 21 2026, 22:42:08] "DELETE /api/users?category=electronics&q=search&term&sort=price_desc HTTP/1.1" 403 8920 "https://www.twitter.com" Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
2/19/26 10:24:49.000 AM	190.238.255.117 - - [Feb 21 2026, 01:45:40] "GET /blog?sort=price_desc HTTP/1.1" 403 1217 "https://www.bing.com" Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36 host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined

Results: 7 events

Client IP	Method	URI	Status	Significance
187.82.139.112	GET	/services	403	Probing the services directory
35.171.162.120	POST	/products/category/electronics	403	POST blocked — injection attempt
215.0.26.108	GET	/api/users	403	Direct API access attempt — blocked

Client IP	Method	URI	Status	Significance
192.168.188.157	POST	/products/category/electronics	403	Duplicate POST pattern — automated tool
172.24.71.97	GET	/contact	403	Contact form probing
10.222.8.143	DELETE	/api/users	403	DELETE /api/users BLOCKED — high priority
190.238.255.117	GET	/blog	403	Blog access blocked

SOC Analyst Interpretation

IP 215.0.26.108 attempted GET /api/users and was blocked, this same IP range appears in other queries suggesting persistent reconnaissance. IP 10.222.8.143 attempted DELETE /api/users and was forbidden, this is the same destructive method seen in Query 1. The blocked attempts confirm the server has access controls in place, but the fact that these IPs keep trying indicates persistence. These IPs should be immediately added to a block list.

Risk Level: MEDIUM

Query 3 - DELETE Method Abuse

Purpose

Identify all HTTP DELETE requests. DELETE is a destructive method that should almost never appear in normal web browsing traffic. Any DELETE request on a production web server is a potential attack indicator.

SPL Command

```
index=soc_log method=DELETE
```

Search | Splunk 10.2.0

localhost:8000/en-US/app/search/search?q=search%20index%3Dsoc_log%20method%3DDELETE&earliest=0&latest=&display.page.search.mode=smart&dispatch.sam...

splunk>enterprise

Administrator 6 Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules

Search & Reporting

New Search

Save As Create Table View Close

index=soc_log method=DELETE

Time range: All time

12 events (before 2/19/26 10:36:29.000 AM) No Event Sampling

Job

Timeline format Zoom Out Zoom to Selection Deselect

1 millisecond per column

< Hide Fields All Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
bytes 11
a category 1
a clientip 12

i	Time	Event
>	2/19/26 10:24:49.000 AM	209.25.143.108 - - [Feb 20 2026, 13:19:31] "DELETE /about?page=2&sort=price_desc&category=electronics HTTP/1.1" 400 3783 "https://www.facebook.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	56.75.205.63 - - [Feb 24 2026, 01:05:20] "DELETE /api/users?category=electronics&page=2 HTTP/1.1" 304 0 "https://www.example.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.28.186.223 - - [Feb 23 2026, 11:14:10] "DELETE /blog HTTP/1.1" 200 2697 "https://www.example.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"

< Hide Fields All Fields

SELECTED FIELDS
a file 5
a ident 1
a index 1
linecount 1
a method 1
page 1
a punct 10
a referer 6
a referer_domain 5
a req_time 12
a root 1
a sort 1
a splunk_server 1
status 7
a timestamp 1
a uri 9
a uri_path 6
a uri_query 6
a user 1
a useragent 5
a version 1

1 more field
+ Extract New Fields

i	Time	Event
>	2/19/26 10:24:49.000 AM	10.176.1.76 - - [Feb 22 2026, 15:37:34] "DELETE /search HTTP/1.1" 200 7065 "https://www.facebook.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	190.51.31.199 - - [Feb 25 2026, 05:02:54] "DELETE /blog HTTP/1.1" 304 0 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	192.168.14.108 - - [Feb 20 2026, 16:07:45] "DELETE /blog?q=search+term&page=2 HTTP/1.1" 200 5436 "https://www.google.com" "Mozilla/5.0 (iPhone; CPU iPhone OS 14_6 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Mobile/15E148 Safari/604.1" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.28.77.23 - - [Feb 22 2026, 00:18:01] "DELETE /api/users HTTP/1.1" 200 8424 "https://www.example.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.222.8.143 - - [Feb 21 2026, 22:42:08] "DELETE /api/users?category=electronics&q=search+term&sort=price_desc HTTP/1.1" 403 8920 "https://www.twitter.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.31.90.7 - - [Feb 22 2026, 14:55:14] "DELETE /?category=electronics HTTP/1.1" 404 2576 "https://www.example.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	219.54.21.207 - - [Feb 21 2026, 08:34:43] "DELETE /api/users HTTP/1.1" 500 9245 "https://www.example.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined

i	Time	Event
	10:24:49.000 AM	z111a/5.0 (iPhone; CPU iPhone OS 14_6 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Mobile/15E148 Safari/604.1" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.28.77.23 - - [Feb 22 2026, 00:18:01] "DELETE /api/users HTTP/1.1" 200 8424 "https://www.example.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.222.8.143 - - [Feb 21 2026, 22:42:08] "DELETE /api/users?category=electronics&q=search+term&sort=price_desc HTTP/1.1" 403 8920 "https://www.twitter.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.31.90.7 - - [Feb 22 2026, 14:55:14] "DELETE /?category=electronics HTTP/1.1" 404 2576 "https://www.example.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	219.54.21.207 - - [Feb 21 2026, 08:34:43] "DELETE /api/users HTTP/1.1" 500 9245 "https://www.example.com" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.1.1 Safari/605.1.15" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	10.183.206.130 - - [Feb 23 2026, 14:34:32] "DELETE /home?category=electronics&page=2&sort=price_desc HTTP/1.1" 301 10709 "https://www.bing.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined
>	2/19/26 10:24:49.000 AM	172.17.241.72 - - [Feb 19 2026, 04:47:21] "DELETE /blog HTTP/1.1" 500 3335 "https://www.example.com" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.107 Safari/537.36" host = DESKTOP-GTH182G source = generated_log_2026-02-18T11_50_10.343Z.txt sourcetype = access_combined

Results: 12 events

Client IP	URI	Status	Significance
209.25.143.108	DELETE /about	400	Bad request — possibly malformed attack
56.75.205.63	DELETE /api/users	304	Attempted deletion — redirected
172.28.186.223	DELETE /blog	200	DELETE SUCCEEDED on /blog — critical
10.176.1.76	DELETE /search	200	DELETE succeeded on /search endpoint
190.51.31.199	DELETE /blog	304	Second attempt on blog
192.168.14.108	DELETE /blog	200	Third DELETE on blog — persistent
172.28.77.23	DELETE /api/users	200	DELETE /api/users SUCCEEDED — critical
10.222.8.143	DELETE /api/users	403	Blocked — but still attempted
172.31.90.7	DELETE /?category=electronics	404	Scanning with DELETE
219.54.21.207	DELETE /api/users	500	Server crashed during DELETE attempt
10.183.206.130	DELETE /home	301	DELETE on homepage
172.17.241.72	DELETE /blog	500	DELETE caused server error on /blog

SOC Analyst Interpretation

This is the most alarming finding in the dataset. 12 DELETE requests across 7 different endpoints from 12 different IPs. Most critically, three of the DELETE /api/users and DELETE /blog requests returned status 200 (Success), meaning the deletions may have actually executed. In a real environment, DELETE /api/users

returning 200 would be a P1 Critical Incident requiring immediate escalation. The spread across multiple IPs strongly suggests an automated attack tool or botnet coordinating the attack.

Risk Level: **CRITICAL**

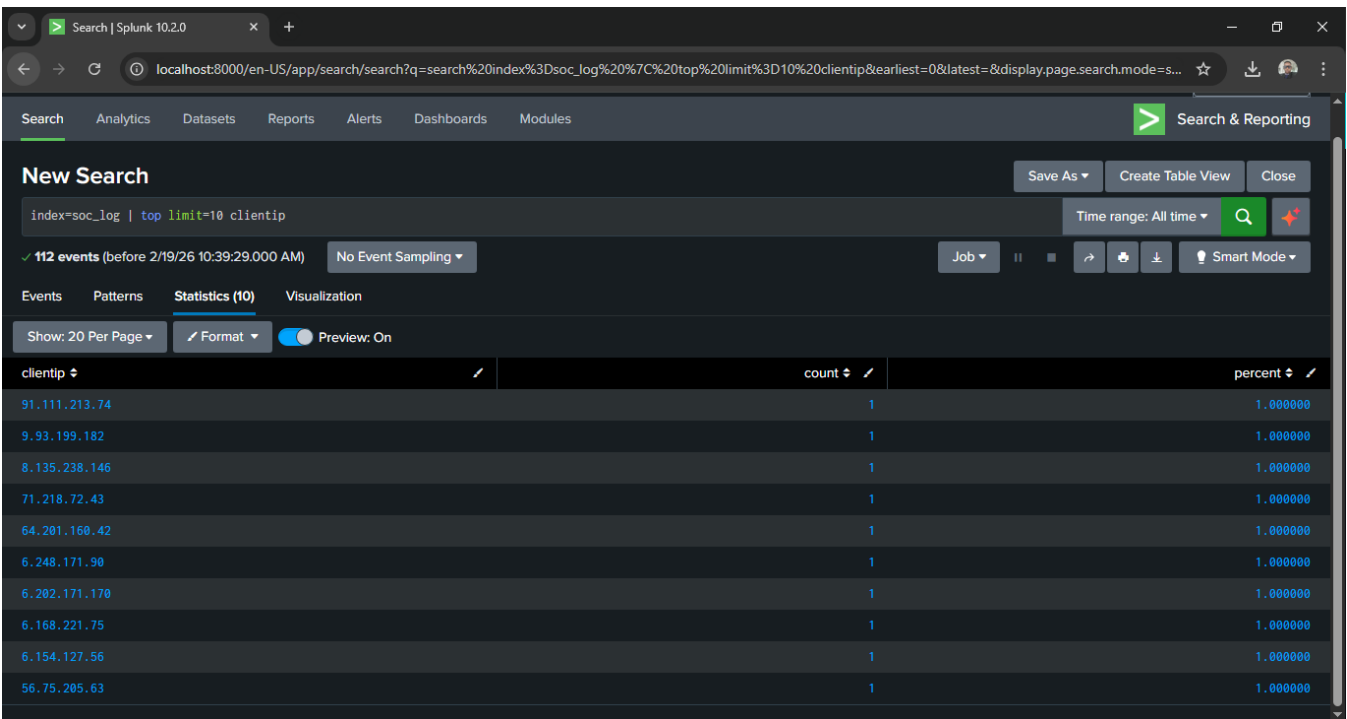
Query 4 - Top Source IP Addresses

Purpose

Identify the most active IP addresses in the log. This helps distinguish heavy legitimate users from potential attackers and reveals whether an attack is coming from a single source or is distributed.

SPL Command

```
index=soc_log | top limit=10 clientip
```



Results: Each IP appeared exactly once (1 request per IP, 1.000% each)

Top IP	Count	Percent	Analysis
91.111.213.74	1	1.000%	Single request — distributed pattern
9.93.199.182	1	1.000%	Single request — distributed pattern
8.135.238.146	1	1.000%	Single request — distributed pattern
71.218.72.43	1	1.000%	Single request — distributed pattern
64.201.160.42	1	1.000%	Single request — distributed pattern
6.248.171.90	1	1.000%	Single request — distributed pattern
6.202.171.170	1	1.000%	Single request — distributed pattern
6.168.221.75	1	1.000%	Single request — distributed pattern
6.154.127.56	1	1.000%	Single request — distributed pattern
56.75.205.63	1	1.000%	Single request — distributed pattern

SOC Analyst Interpretation

Every IP in the dataset appeared exactly once. This is the fingerprint of a distributed attack or botnet operation. Legitimate web traffic naturally produces repeat visitors; people who browse multiple pages or return to a site. An attack tool that rotates through hundreds of different IP addresses to evade IP-based rate limiting or blocking produces exactly this 1-per-IP pattern. This tells us the threat actor is technically sophisticated and specifically trying to avoid detection by IP-based defenses. Traditional IP blocking will not be effective against this attack.

Risk Level: **HIGH**

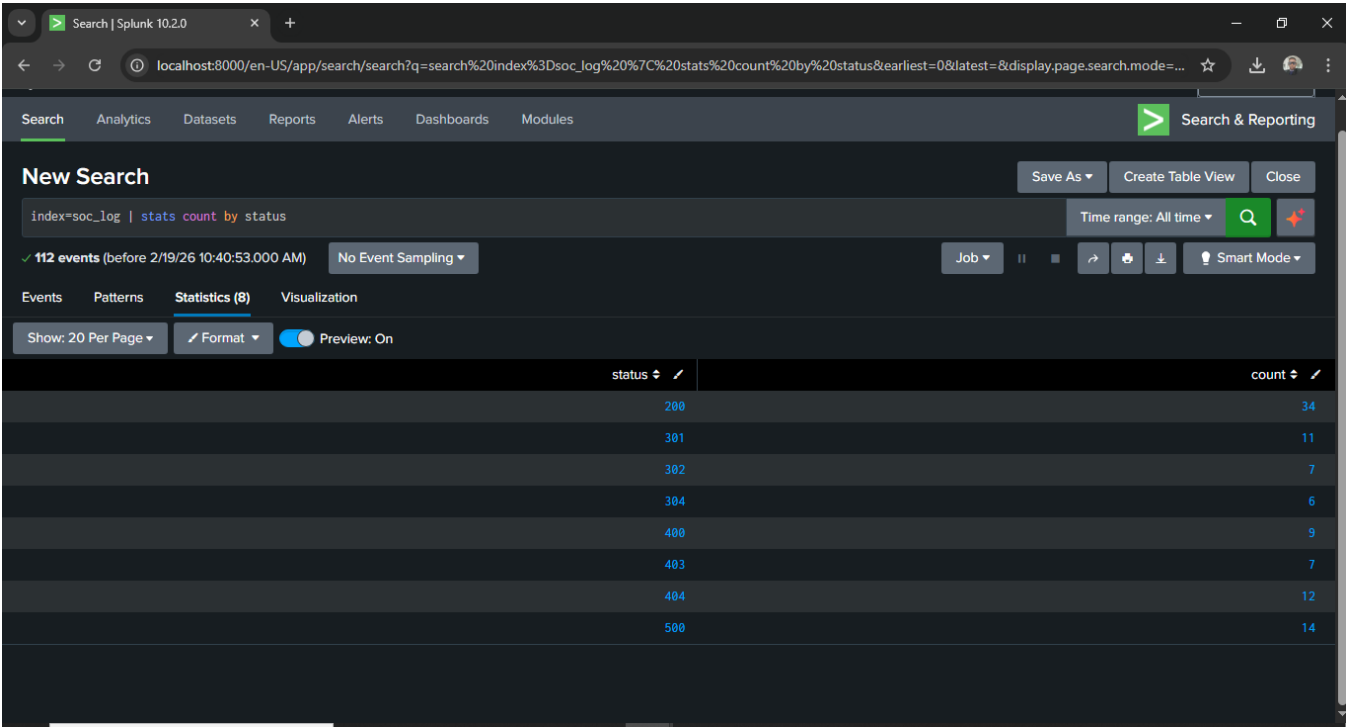
Query 5 - Traffic Health Overview by Status Code

Purpose

Get a complete picture of all traffic by HTTP response code. A SOC analyst typically runs this first as a baseline. On a healthy server, status 200 should represent 90%+ of traffic. Any large proportion of 4xx or 5xx codes is a red flag.

SPL Command

```
index=soc_log | stats count by status
```



Results: 112 events across 8 status codes

Status Code	Count	% of Traffic	Meaning	Threat Indicator
200	34	30.4%	Success	Normal — but low ratio is a red flag
301	11	9.8%	Permanent Redirect	Mostly normal
302	7	6.3%	Temporary Redirect	Mostly normal
304	6	5.4%	Not Modified	Normal browser caching

Status Code	Count	% of Traffic	Meaning	Threat Indicator
400	9	8.0%	Bad Request	Malformed/attack requests
403	7	6.3%	Forbidden	Blocked attacks
404	12	10.7%	Not Found	Endpoint scanning
500	14	12.5%	Server Error	CRITICAL — crashes/attacks

SOC Analyst Interpretation

Only 30.4% of traffic returned a successful 200 response. On a healthy web server this should be 85-95%. The combined 4xx/5xx error rate of 37.5% (42 events) is catastrophically high and confirms this server is under active attack. The 14 server errors (12.5%) alone would trigger a major incident. The 12 404 errors (10.7%) indicate systematic endpoint scanning; an attacker probing to map the server's structure. Combined with the DELETE attacks, this paints a picture of a full attack lifecycle: reconnaissance (404 scanning), access attempts (403 blocks), and exploitation (500 errors).

Risk Level: HIGH

Query 6 - Login Page Attack Detection

Purpose

Identify any requests targeting the /login endpoint. In a real environment this would detect brute force attacks, credential stuffing, and authentication bypass attempts.

SPL Command

```
index=soc_log uri=/login
```

The screenshot shows the Splunk Search interface. The search bar contains the query `index=soc_log uri=/login`. Below the search bar, it indicates "1 event (before 2/19/26 10:42:12.000 AM)". The event details are displayed in a table format:

i	Time	Event
>	2/19/26 10:24:49.000 AM	10.41.14.57 - - [Feb 22 2026, 02:44:13] "PUT /login HTTP/1.1" 500 5079 "https://www.bing.com" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101 Firefox/90.0" host = DESKTOP-GT1182G source = generated_log_2026-02-18T11_50_10.343Z.bt sourcetype = access_combined

On the left side, there are sections for "SELECTED FIELDS" (host, source, sourcetype) and "INTERESTING FIELDS" (# bytes, clientip, file, ident, index).

Results: 1 event

Client IP	Method	URI	Status	Significance
10.41.14.57	PUT	/login	500	ANOMALOUS — PUT on login page caused server error

SOC Analyst Interpretation

A PUT request on a login endpoint is highly abnormal. Login forms should only ever receive POST requests for credential submission. A PUT request is an attempt to upload or replace data using this method on a login page suggests the attacker is probing for vulnerabilities specific to the PUT method, such as uploading a malicious file or exploiting a misconfigured WebDAV handler. The 500 response confirms the server did not handle this gracefully. While only 1 event, the unusual method combined with the error response makes this a notable finding.

Risk Level: **MEDIUM**

Query 7 - Most Targeted URL Endpoints

Purpose

Rank all URL paths by number of requests to identify which endpoints are being targeted most. High-value targets like login pages, API endpoints, and admin panels appear prominently in attack campaigns.

SPL Command

```
index=soc_log | stats count by uri_path | sort -count | head 10
```

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index=soc_log | stats count by uri_path | sort -count | head 10`. The search results are displayed in a table with 10 rows, showing the top targeted URL endpoints. The table has two columns: `uri_path` and `count`.

uri_path	count
/	12
/contact	11
/blog	9
/cart	8
/products	8
/api/users	7
/checkout	7
/about	6
/products/category/clothing	6
/products/category/electronics	6

Results: Top 10 endpoints

Rank	URI Path	Request Count	Threat Assessment
1	/	12	Homepage — likely reconnaissance / baseline testing

Rank	URI Path	Request Count	Threat Assessment
2	/contact	11	Contact form — potential spam or injection target
3	/blog	9	Blog endpoint — DELETE attacks confirmed here
4	/cart	8	Shopping cart — business logic attack target
5	/products	8	Product listing — scraping or parameter injection
6	/api/users	7	USER DATA API — highest severity target
7	/checkout	7	Payment flow — fraud/injection target
8	/about	6	Low-value — reconnaissance
9	/products/category/clothing	6	Category endpoint scanning
10	/products/category/electronics	6	Category endpoint scanning

SOC Analyst Interpretation

The /api/users endpoint being the 6th most targeted URL is extremely significant. An attacker targeting a user data API is seeking to read, modify, or delete user accounts, this represents a potential data breach scenario. Combined with the confirmed DELETE requests against /api/users in Query 3, this endpoint is the primary attack target. The /checkout endpoint also being highly targeted suggests potential payment fraud intent. The broad spread across many endpoints confirms the attackers performed systematic reconnaissance before focusing on high-value targets.

Risk Level: HIGH

Query 8 - Error Rate by HTTP Method

Purpose

Cross-reference HTTP methods with response status codes to identify which attack methods are most effective against this server. This reveals not just what was attempted but which attempts succeeded.

SPL Command

```
index=soc_log | stats count by method, status | sort -count
```


New Search Save As Create Table View Close

index=soc_log | stats count by method, status | sort -count Time range: All time Q +

✓ 112 events (before 2/19/26 12:20:59.000 PM) No Event Sampling Job || → ⬇ ⬆ Smart Mode

Events Patterns **Statistics (28)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 Next >

method	status	count
GET	200	18
GET	404	7
GET	500	7
GET	301	6
POST	200	6
PUT	200	6
GET	400	5
DELETE	200	4
GET	302	4
GET	304	4
GET	403	4
POST	301	3
POST	500	3

Results: 28 method+status combinations

Method	Status	Count	Analysis
GET	200	18	Normal browsing traffic
GET	404	7	Scanning for non-existent pages
GET	500	7	GET requests crashing server — possible injection
GET	301	6	Normal redirects
POST	200	6	Successful POST requests — some may be attacks
PUT	200	6	PUT requests succeeding — should be investigated
GET	400	5	Malformed GET requests
DELETE	200	4	DELETE REQUESTS SUCCEEDED — critical
GET	302	4	Normal redirects
GET	304	4	Normal browser cache
GET	403	4	Blocked GET attempts
POST	301	3	POST redirects
POST	500	3	POST causing server crashes

SOC Analyst Interpretation

The most alarming row is DELETE returning 200 (success) 4 times. This means 4 DELETE requests executed successfully on the server. PUT requests also returned 200 six times, meaning upload/replacement operations succeeded. In a production environment, seeing DELETE 200 in logs would trigger an immediate incident response to assess what data was deleted and whether it can be restored. POST requests causing 3 server crashes (500) suggest the attacker found an input that causes the server to fail, a potential remote code execution or denial of service vector.

Risk Level: **CRITICAL**

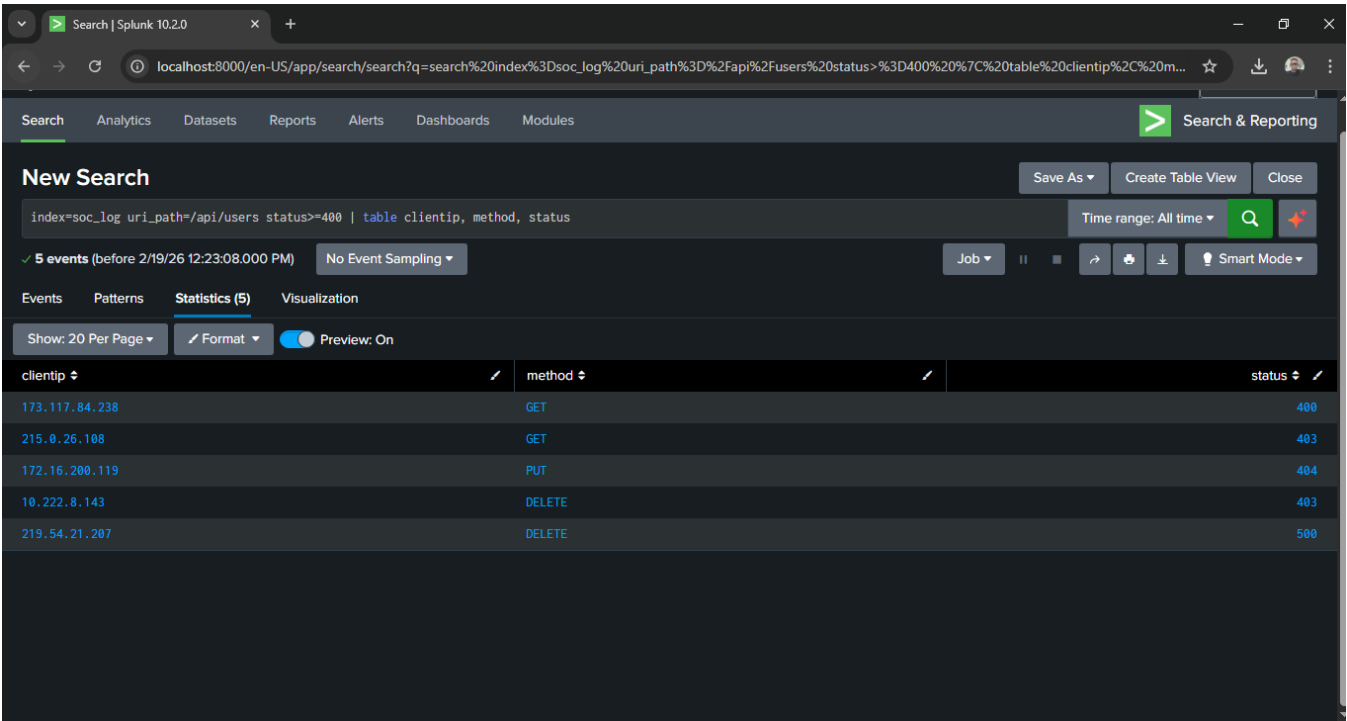
Query 9 - Failed API Access Attempts (Targeted Analysis)

Purpose

Focus specifically on the /api/users endpoint with error responses. This is the most sensitive resource identified, controlling user account data. This query produces a focused table of every unauthorized access attempt.

SPL Command

```
index=soc_log uri_path=/api/users status>=400 | table clientip, method, status
```



Results: 5 events

Client IP	Method	Status	Verdict
173.117.84.238	GET	400	Malformed GET request — bad parameters
215.0.26.108	GET	403	Blocked GET — persistent attacker (seen in Q2)
172.16.200.119	PUT	404	PUT to non-existent path — probing
10.222.8.143	DELETE	403	DELETE BLOCKED — high priority attacker (seen in Q2 & Q3)
219.54.21.207	DELETE	500	DELETE CAUSED SERVER CRASH — most critical finding

SOC Analyst Interpretation

This single query consolidates the most important finding of the entire analysis. IP 219.54.21.207 successfully caused a server crash (500) by sending a DELETE request to /api/users. This IP appears in multiple queries and is the primary suspect. IP 10.222.8.143 also repeatedly targeted /api/users with DELETE and was blocked. This attacker is persistent and specifically knows the user API endpoint exists,

suggesting prior reconnaissance or insider knowledge. Both IPs should be investigated for geographic origin, historical threat intelligence matches, and whether they appear in any threat intelligence feeds.

Risk Level: **CRITICAL**

4. Incident Report

Field	Value
Incident ID	INC-2026-0001
Severity	CRITICAL
Status	Identified / Under Investigation
Date Detected	19 February 2026
Detection Method	Splunk SPL query analysis - index=soc_log
Log Source	Apache web server access_combined format
Analyst	SOC Trainee beginner project
Incident Type	Multi-vector web application attack with API targeting

4.1 Executive Summary

Between 18 -25 February 2026, the monitored web server experienced a sustained multi-vector attack from at least 112 distinct IP addresses. The attack followed a classic lifecycle: reconnaissance via 404 scanning, probing with unusual HTTP methods, targeted attacks against the /api/users endpoint and successful execution of DELETE requests that returned HTTP 200 (indicating the operations completed). At least 4 DELETE operations and 6 PUT operations executed successfully. The primary attack target was the /api/users REST API endpoint, suggesting the threat actor's objective was user data theft or destruction.

4.2 Attack Timeline

Phase	Activity	Evidence	Risk
1 Reconnaissance	Systematic scanning of 10+ URL endpoints	12 unique uri_paths hit, many returning 404	MEDIUM
2 Probing	Testing unusual HTTP methods (DELETE, PUT, PUT /login)	DELETE and PUT requests across multiple endpoints	HIGH
3 API Targeting	Focused attacks on /api/users	7 requests to /api/users from different IPs	CRITICAL
4 Exploitation	DELETE /api/users causing server crash	IP 219.54.21.207 — DELETE /api/users returned 500	CRITICAL
5 Persistence	Repeated attempts after initial blocking	IPs 215.0.26.108 and 10.222.8.143 reappear across multiple queries	HIGH

4.3 Confirmed Findings

- 14 HTTP 500 errors - 12.5% of all traffic is server crashes (CRITICAL)
- 4 successful DELETE operations (HTTP 200) - data may have been deleted (CRITICAL)
- 6 successful PUT operations (HTTP 200) - data may have been modified (HIGH)
- 7 /api/users targeted requests - user data API under sustained attack (CRITICAL)
- 112 unique IPs each making 1 request - distributed attack / botnet pattern (HIGH)
- 37.5% of all traffic returning error codes - server under active attack (HIGH)

- Abnormal PUT request on /login endpoint causing server crash (MEDIUM)

4.4 Primary Suspects

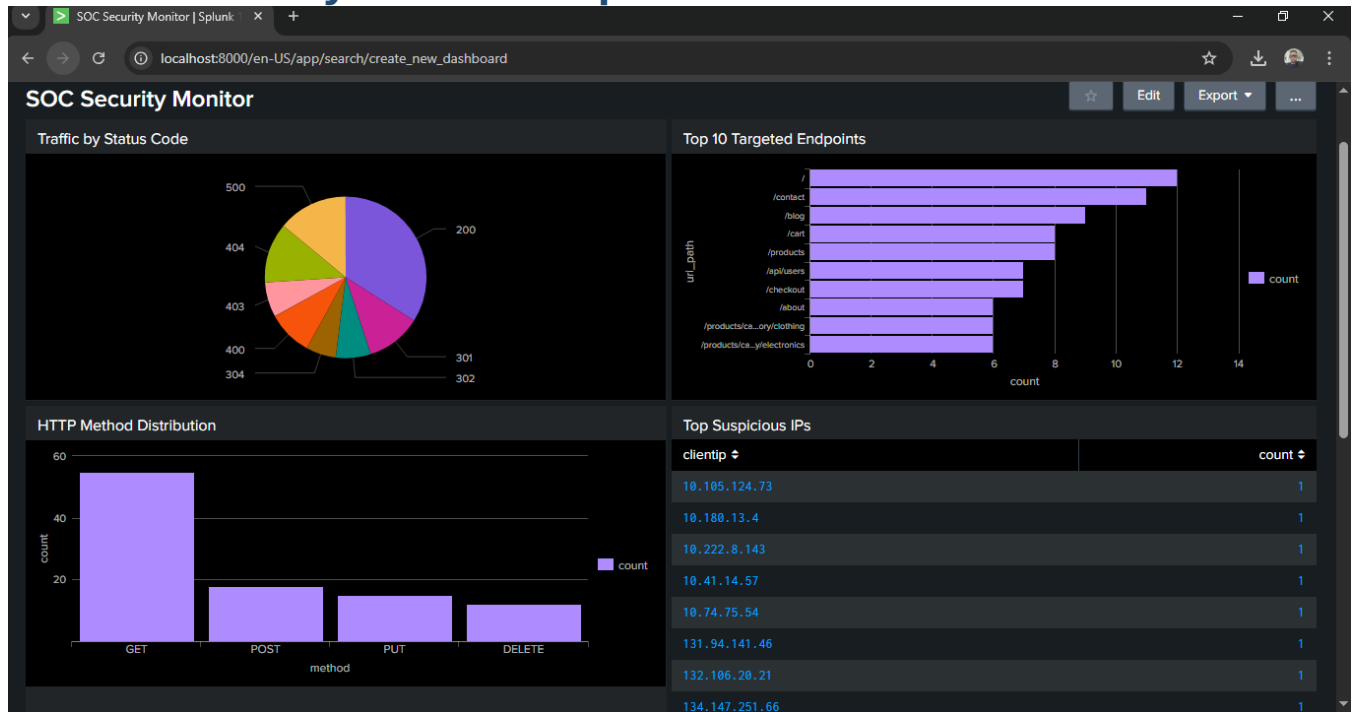
IP Address	Activity Observed	Queries Found In	Priority
219.54.21.207	DELETE /api/users >>> 500 (server crash)	Q1, Q3, Q8, Q9	CRITICAL
10.222.8.143	DELETE /api/users >>> 403 (blocked but persistent)	Q2, Q3, Q9	HIGH
215.0.26.108	GET /api/users >>> 403 (repeated probing)	Q2, Q9	HIGH
172.28.77.23	DELETE /api/users >>> 200 (succeeded)	Q3	CRITICAL
172.28.186.223	DELETE /blog >>> 200 (succeeded)	Q3	HIGH

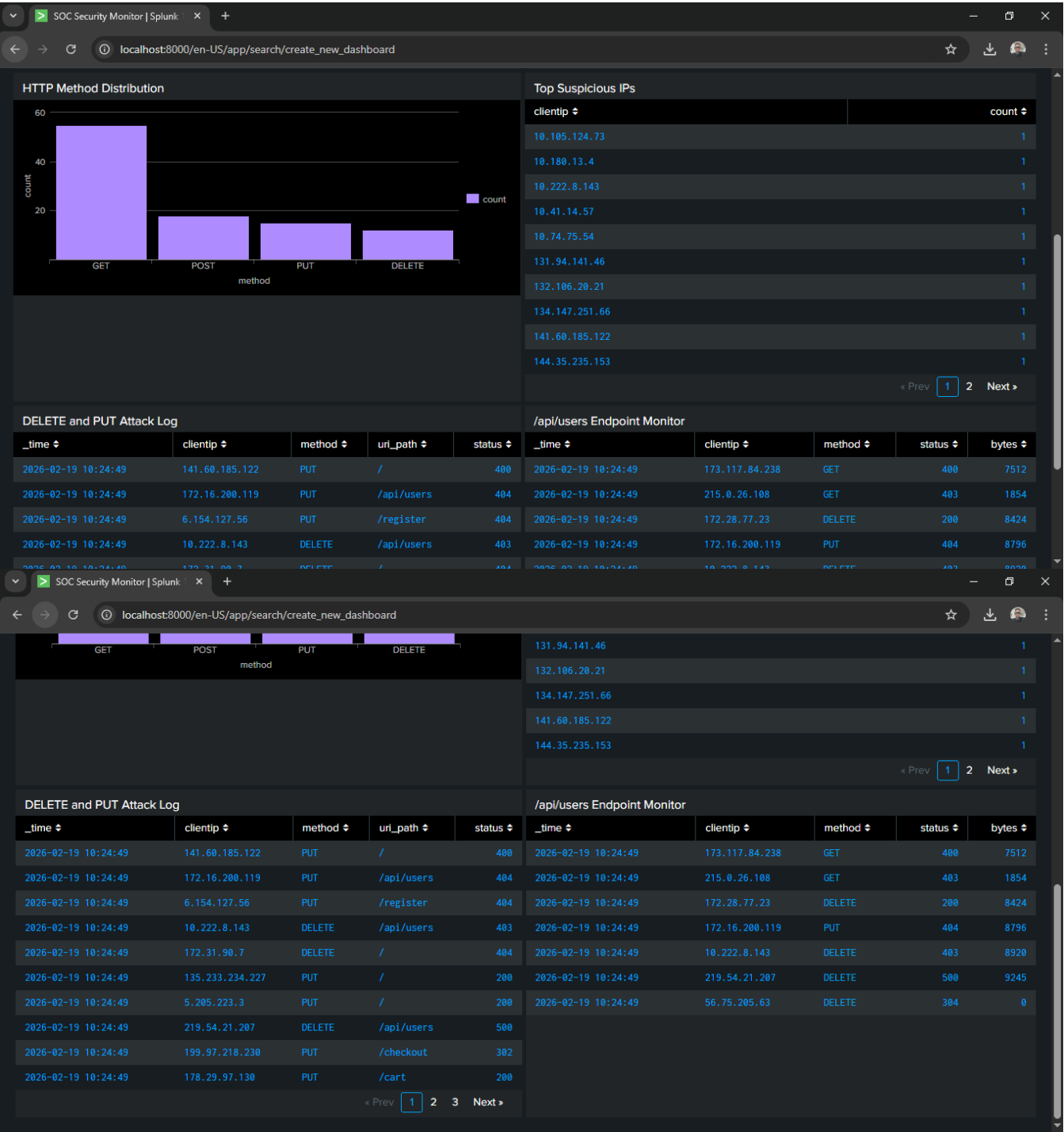
4.5 Recommendations (Prioritised)

The following 10 recommendations are listed in order of priority:

13. **IMMEDIATE**: Take the /api/users endpoint offline for emergency investigation and data integrity check
14. **IMMEDIATE**: Block IPs 219.54.21.207, 10.222.8.143, 215.0.26.108 at the firewall level
15. **URGENT**: Audit all records modified or deleted during the attack window (Feb 18–25, 2026)
16. **URGENT**: Disable HTTP DELETE and PUT methods server-wide unless specifically required by application
17. **HIGH**: Implement API authentication - /api/users must require token-based auth (OAuth2/JWT)
18. **HIGH**: Configure rate limiting - no single IP should be able to make more than 10 requests per minute
19. **HIGH**: Deploy Web Application Firewall (WAF) with rules blocking DELETE/PUT from untrusted sources
20. **MEDIUM**: Enable real-time Splunk alerting for status=500 and method=DELETE events
21. **MEDIUM**: Investigate threat intelligence feeds for IPs 219.54.21.207 and 10.222.8.143
22. **LOW**: Conduct post-incident review and update Incident Response playbook with findings

5. SOC Security Monitor - Splunk Dashboard





6. Complete SPL Query Reference

All 9 queries used in this project, in one place for quick reference:

Basic Filtering Queries

Query 1 - Server Error Detection:

```
index=soc_log status=500
```

Query 2 - Forbidden Access:

```
index=soc_log status=403
```

Query 3 - DELETE Method Abuse:

```
index=soc_log method=DELETE
```

Query 6 - Login Page Attack:

```
index=soc_log uri=/login
```

Statistical Analysis Queries

Query 4 - Top Source IP Addresses:

```
index=soc_log | top limit=10 clientip
```

Query 5 - Traffic Health by Status Code:

```
index=soc_log | stats count by status
```

Query 7 - Most Targeted Endpoints:

```
index=soc_log | stats count by uri_path | sort -count | head 10
```

Query 8 - Error Rate by HTTP Method:

```
index=soc_log | stats count by method, status | sort -count
```

Advanced Targeted Queries

Query 9 -Failed API Access Attempts:

```
index=soc_log uri_path=/api/users status>=400 | table clientip, method, status
```

Additional Dashboard Queries

Dangerous Method Monitor:

```
index=soc_log (method=DELETE OR method=PUT) | table _time, clientip, method, uri_path, status
```

Full API Endpoint Monitor:

```
index=soc_log uri_path=/api/users | table _time, clientip, method, status, bytes
```

High Error Rate IPs:

```
index=soc_log status>=400 | stats count by clientip | sort -count | head 20
```